

B.Tech II Year I Semester (R20) Supplementary Examinations November 2024

SIGNALS & SYSTEMS

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Define a linear system. | 2M |
| (b) What is the difference between continuous-time and discrete-time signals? | 2M |
| (c) Define the Fourier transform of a continuous-time signal. | 2M |
| (d) What is the Nyquist rate? | 2M |
| (e) Define the Laplace transform of a signal. | 2M |
| (f) Write the Laplace transform of the unit step function. | 2M |
| (g) Define LTI system. | 2M |
| (h) What is rise time? | 2M |
| (i) Difference between Fourier transform and discrete-time Fourier transform. | 2M |
| (j) Write the Z-transform of $x[n] = [1, 2, 3]$. | 2M |

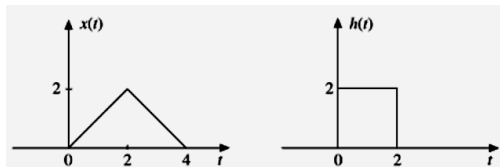
PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Illustrate the operations performed on continuous-time signals with suitable examples. 5M
- (b) Determine the even and odd components of the signal 5M
- $$x(t) = \cos(t) + \sin(t) + \cos(t) \sin(t).$$

OR

- 3 Find the convolution of the signal $x(t)$ and $h(t)$ given below. 10M



- 4 (a) State and prove the timing-shifting and frequency-shifting properties of the Fourier transform. 5M
- (b) Obtain the Fourier Transform of the two-sided exponential signal 5M
- $$x(t) = e^{-a|t|}.$$

OR

- 5 (a) Explain the sampling theorem and discuss aliasing with necessary diagrams. 5M
- (b) Obtain the exponential Fourier series coefficients of the signal 5M

$$x(t) = 1 + \sin(\omega_0 t) + 2 \cos(\omega_0 t) + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

- 6 (a) List two properties of ROC of Laplace transform. 5M
- (b) Draw the S-plane of the system and comment on its stability for transfer function 5M
- $$H(s) = \frac{s+1}{s^2+3s+2}.$$

OR

Contd. in Page 2

- 7 (a) Describe how the Laplace transform is used to solve linear differential equations. 5M
(b) Find the Inverse Laplace Transform of $X(s) = \frac{(s+4)}{(s+3)(s+5)}$. 5M
- 8 (a) Describe distortion less transmission through the system. 5M
(b) Discuss the system properties like causality and BIBO stability. Test the given continuous time system, $y(t) = x(2t)$ for causality and stability. 5M
- OR**
- 9 (a) Explain the filter characteristics of linear systems. 5M
(b) Discuss Causality and Paley-Wiener criterion for physical realization systems. 5M
- 10 (a) State and prove the initial value theorem for Z-transforms. 5M
(b) Find the Z-Transform of the discrete-time sequence $x[n] = (0.5)^n u[n]$. 5M
- OR**
- 11 (a) What is the significance of solving difference equations using the Z-transform? 5M
(b) Discuss the BIBO stab. 5M

B.Tech II Year I Semester (R20) Supplementary Examinations August/September 2023

SIGNALS & SYSEMS

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|---|----|
| (a) Define Correlation of two signals. | 2M |
| (b) Is the discrete in time signal $\sin(31\pi n)$ is periodic or aperiodic. | 2M |
| (c) When Aliasing Phenomenon is occurred and how it can be eliminated? | 2M |
| (d) Write the Modulation property of Fourier Transform. | 2M |
| (e) Given impulse reponse of a system $h(t) = e^{-2t}u(t)$, check whether the system is stable or not. | 2M |
| (f) Find the ROC of $h(t) = e^{-2t}u(t) + e^t u(t)$. | 2M |
| (g) Define System Bandwidth. | 2M |
| (h) Write the condition for a LTI system to be a distortion less System. | 2M |
| (i) Find the Z-Transform of $h(n) = a^{-2n}u(n)$. | 2M |
| (j) Find the Final value of $h(n) = \sin(31\pi n)u(n)$. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | |
|---|----|
| 2 (a) Determine whether the following function is periodic or not. If so find the period.
$x(t) = 3\sin 200\pi t + 4\cos 100t$. | 5M |
| (b) Define mean square error and derive the expression for evaluating mean square error. | 5M |
| OR | |
| 3 (a) Define the error function while approximating signals and hence derive the expression for condition for orthogonality between two waveforms $f_1(t)$ and $f_2(t)$. | 5M |
| (b) Explain the properties of unit impulse function. | 5M |
| 4 (a) State and prove Sampling Theorem for Low pass Signals. | 5M |
| (b) Find the Fourier transform of signum function. | 5M |
| OR | |
| 5 (a) What is the difference between Fourier series Analysis and Fourier Transforms? Explain with an example. | 5M |
| (b) State and Prove Modulation property of Fourier Transform. | 5M |
| 6 (a) Determine the Laplace transform of
$x(t) = e^{-at}u(t) + e^{-bt}u(t)$ and identify the ROC ,poles and zeros located in s-plane. | 5M |
| (b) Find the output response of the system described by a differential equation
$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = x(t)$ when the input signal $x(t) = \cos(t)$. The initial conditions are
$\frac{dy(0^+)}{dt} = 1$ $y(0^+) = 1$. | 5M |
| OR | |
| 7 (a) State and prove all the properties of Region of Convergence of Laplace Transform. | 5M |
| (b) Find the inverse Laplace transform of $X(s) = \frac{s+2}{s^3+7s^2+15s+9}$. | 5M |

Contd. In Page 2

- 8 (a) Derive the relation between the bandwidth and rise time of a LTI system. 5M
(b) Given a signal $x(t) = e^{-2t}u(t)$; impulse response of LTI system. $h(t) = e^{-2t}u(t)$, find output $Y(\omega)$. 5M

OR

- 9 (a) What is Impulse Response? Show that the response of an LTI system is convolution integral of its impulse Response with input signal? 5M
(b) Define Energy and Power spectral densities and give their Physical significance. 5M
- 10 (a) Find the z-transform of $x(n) = a^n \sin(\omega n) u(n)$. 5M
(b) Find inverse Z-Transform of $X(Z) = (1 - z^{-1})/(z - 1/2)(z - 1/3)$; for ROC $|Z| > 1/2$. 5M

OR

- 11 (a) Obtain the Z-transform of $x(n) = -a^n u(-n - 1)$ and find its ROC. 5M
(b) State and explain the following properties of Z-transform. 5M
(i) Convolution (ii) Time Shifting (iii) Time Reversal (iv) Frequency Shifting.

B.Tech II Year I Semester (R20) Supplementary Examinations April/May 2024

SIGNALS & SYSTEMS
(Electronics & Communication Engineering)

Time: 3 hours

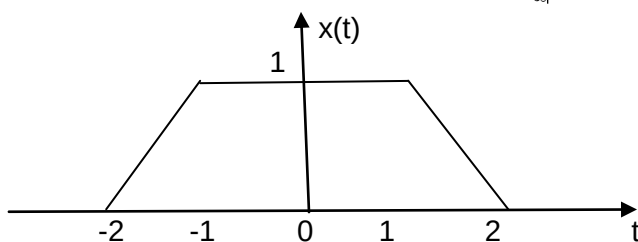
Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) Define Signal and System. 2M
 - (b) Find the time period of a signal, $x(t) = 5 \cos(4t) + 4 \sin(5t)$. 2M
 - (c) Discuss any two symmetry properties of trigonometric Fourier series. 2M
 - (d) Obtain the FT of $\cos(2t)$. 2M
 - (e) Discuss the concept "Existence of Laplace Transform". 2M
 - (f) Obtain the transfer function of the continuous time LTI system governed by the differential equation: $\frac{d^2}{dt^2}y(t) + 4\frac{d}{dt}y(t) + 3y(t) = 2\frac{d}{dt}x(t) + 4x(t)$. Assume, zero initial conditions. 2M
 - (g) When the system is said to be physically realizable? Discuss all the related conditions. 2M
 - (h) Why the system, $y(t) = T\{x(t)\} = 2x(t) + 3$ is linear time invariance? 2M
 - (i) Evaluate the system function in w-domain of a discrete system having LCCDE, $y(n) - \frac{1}{2}y(n-1) = x(n) + \frac{1}{2}x(n-1)$. 2M
 - (j) Determine the z-domain of a unit step sequence, $x(n) = -u(-n-2)$. 2M

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) List the various operations used in continuous time signals and discuss any three with suitable examples. 5M
- (b) Use graphical method to evaluate the integral $\int_{-\infty}^{\infty} \left| \frac{d^2}{dt^2} x(t) \right| dt$ for the signal given below, 5M



OR

- 3 (a) Explain the following with suitable examples: 5M
- (i) Bounded and unbounded discrete signals,
 - (ii) Energy and Power discrete signals.
- (b) Draw the graphical representation and sequence form of a discrete time signal 5M
- $x(n) = r(n+2) - r(n-3) - 5u(n-4)$ and also, Evaluate the summation $\sum_{n=0}^{\infty} x(n)$, where $r(n)$ is unit ramp sequence.

- 4 (a) How to represent a signal by using trigonometric Fourier Series? Explain. 5M
- (b) Evaluate the coefficients of trigonometric Fourier series a_0 , a_n and b_n , given $x(t)$ over the interval $(0, 2\pi)$ 5M

$$x(t) = \begin{cases} 1, & 0 < t < \pi \\ -1, & \pi < t < 2\pi \end{cases}$$

OR

Contd. in Page 2

- 5 (a) State and prove time differentiation property of Fourier Transform. 5M
 (b) Apply the properties of Fourier Transform and obtain the frequency domain of (i) Unit step signal $u(t)$ and (ii) Signum signal $\text{Sgn}(t)$. 5M

- 6 (a) State and prove differentiation in s-domain property of Laplace Transform. 5M
 (b) Evaluate the Laplace Transform of $x(t)$, indicate poles, zeroes, and ROC in s-plane, given $x(t) = 3e^{-2t}u(t) - 2e^{-3t}u(t)$. 5M

OR

- 7 (a) Discuss the LTI system having rational system function $H(s)$ for causality and stability with a suitable example. 5M
 (b) Evaluate the causal signal $x(t)$ from its s-domain representation 5M

$$X(s) = \frac{s+1}{(s+2)(s+3)}.$$

- 8 (a) Compare continuous time systems with discrete time systems. 5M
 (b) Examine the continuous time system for Linearity, Time Invariance and BIBO stability. 5M
 $y(t) = T\{x(t)\} = \frac{d}{dt}x(t).$

OR

- 9 (a) Explain ideal and practical characteristics of LPF, HPF, BPF and BSF. 5M
 (b) Draw RC low pass filter circuit and obtain:
 (i) System Function $H(\omega)$,
 (ii) Magnitude Response,
 (iii) Band Width (BW). 5M

- 10 (a) What is DTFT and IDTFT? State any 3 properties of DTFT. 5M
 (b) Apply the properties of DTFT and find, 5M

$$(i) X(e^{j\omega}) \text{ at } \omega = 0 \quad (ii) \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega \quad (iii) \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

Given $x(n) = \{1, 4, 3, 2, 5, 6, 7, 0, 9\}$ and $\text{DTFT}[x(n)] = X(e^{j\omega})$.

OR

- 11 (a) Define the z-transform of a right sided, left sided and both sided sequences. 5M
 (b) Calculate the initial and final values of a causal sequence $x(n)$ from the z-domain 5M

$$X(z) = \frac{z(z+1)(z+2)}{(z-1)(z-1/2)(z-1/4)(z-1/8)}.$$
